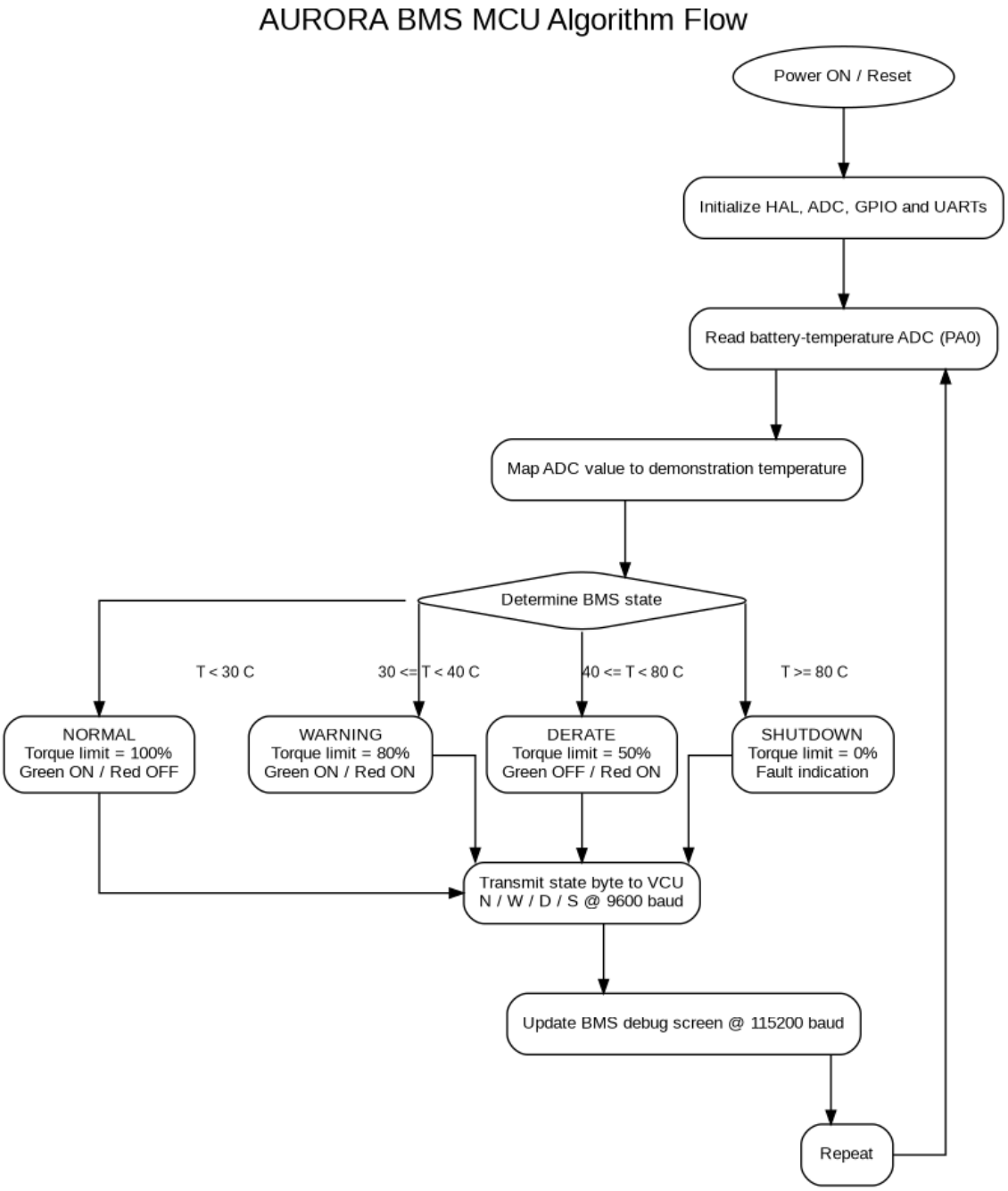
AURORA SDV MCU Flow Diagrams

Embedded algorithms used in the Proteus STM32 prototype

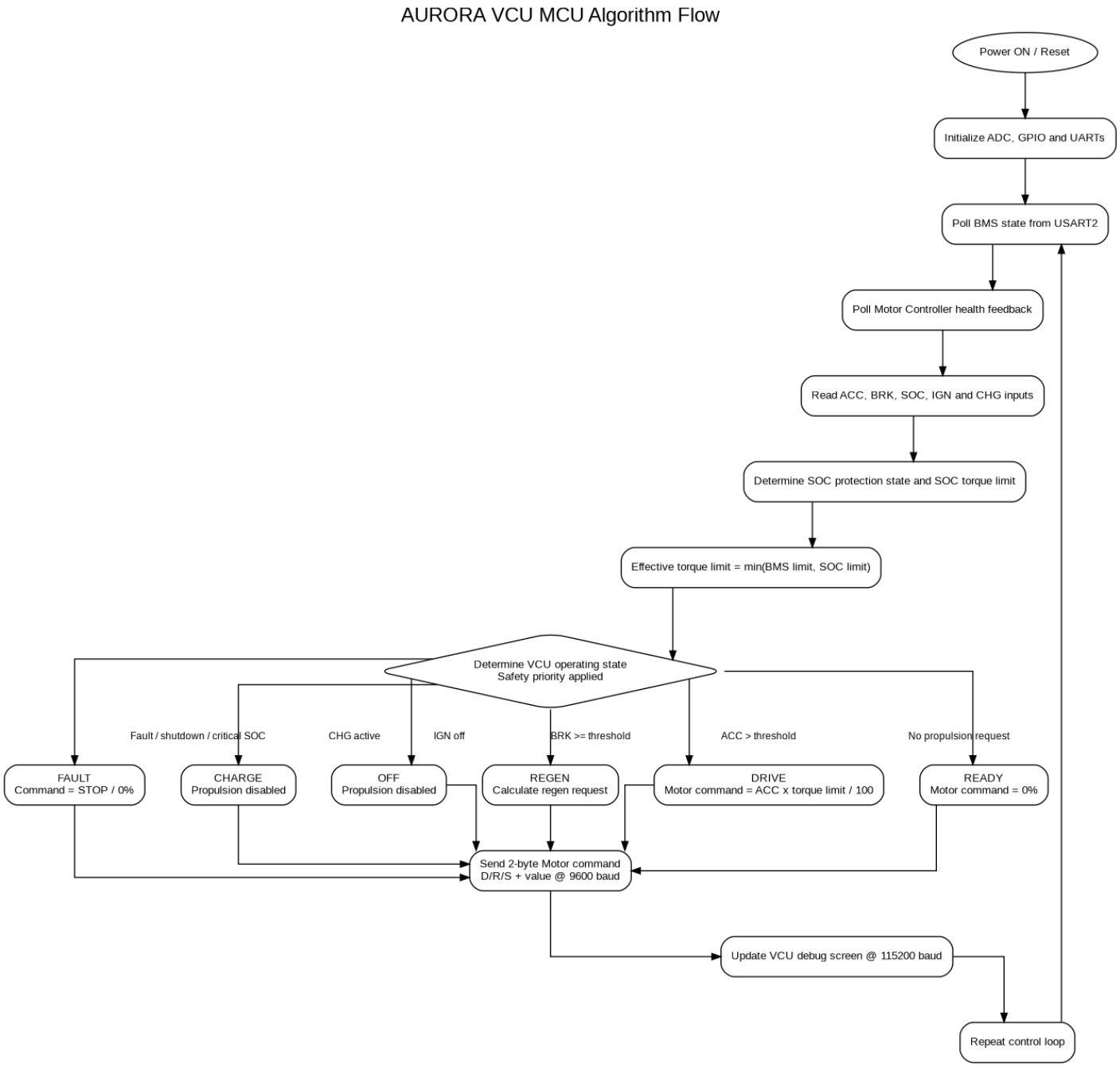
1. BMS MCU Flow

Battery-temperature acquisition, state classification, LED indication, torque-limit generation and BMS-to-VCU transmission.



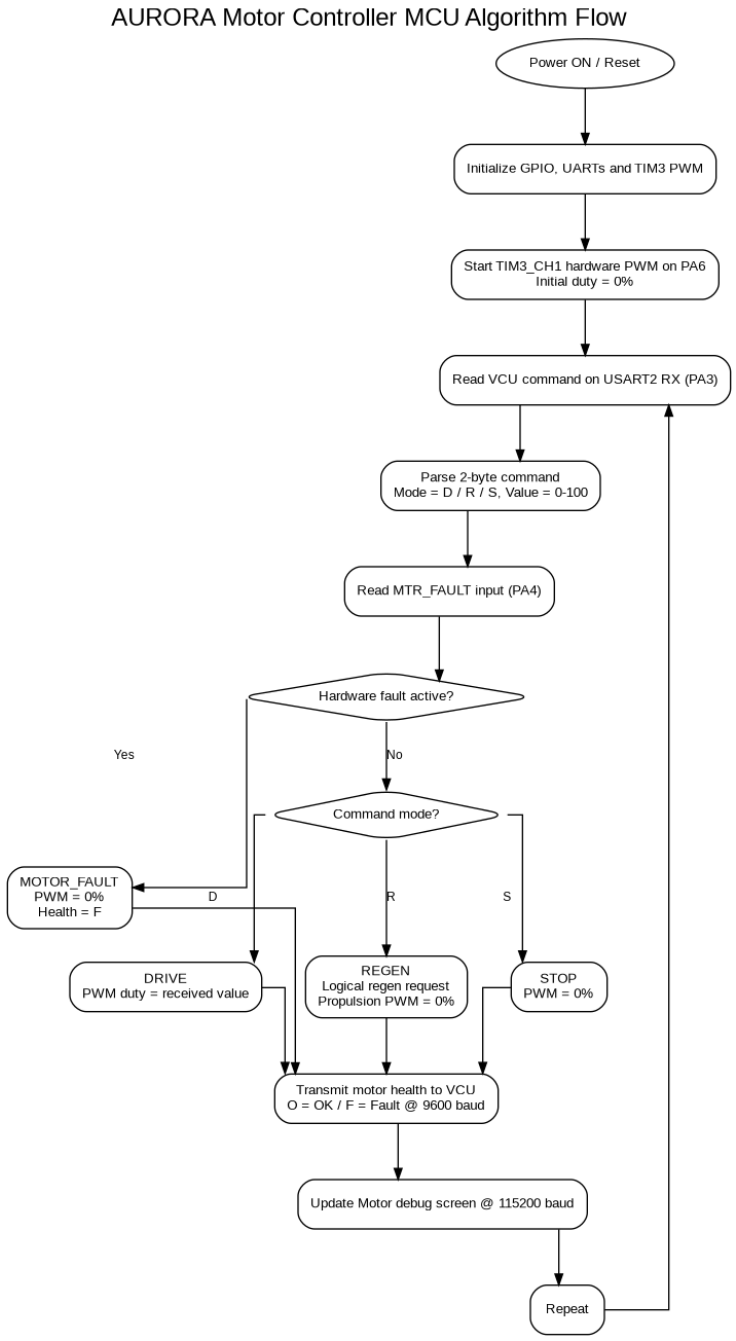
2. VCU MCU Flow

Driver-input acquisition, BMS/SOC protection, state prioritization, torque arbitration and command transmission to the Motor Controller.



3. Motor Controller MCU Flow

VCU command reception, hardware-fault handling, DRIVE/REGEN/STOP processing, TIM3 PWM generation and health feedback.



4. Overall Embedded Control Flow

End-to-end signal flow from battery temperature and driver inputs through the BMS, VCU and Motor Controller to the simplified motor stage.

